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Optimizing Hybrid Fuel Cell Systems for Sustainable Energy Transition

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Abstract

As global energy demand continues to rise, the need for alternative energy also rises. In addition to renewable energy, hydrogen energy has emerged as a promising solution for sustainable energy generation. Since renewable sources such as wind and solar are intermittent, hydrogen can enable renewables to provide an even greater contribution by storing and transporting the energy that comes from them.

This paper explores the technological advancements, and challenges of hydrogen fuel cells, including cost barriers, durability concerns, and infrastructure limitations. A central theme of this review is the role of optimization in overcoming challenges such as high cost, low efficiency and power output, and high losses. The paper reports recent efforts in optimizing flow field geometries, gas diffusion layer (GDL) and catalyst layer (CL) structures, and non-uniform catalyst loading, which collectively enhance electrochemical performance, water management, and cost-effectiveness. It also identifies critical research gaps, such as the scarcity of full-stack modeling, compression effects in GDLs, and limited experimental validation under realistic conditions.

The paper also discusses the potential of hybrid fuel cell systems as a promising approach to overcome the limitations of using solely fossil-based or renewable hydrogen sources. By combining the environmental benefits of renewables with the stability of conventional energy, hybrid systems can enhance reliability and reduce emissions. The importance of optimization across energy inputs, material selection and system design is also highlighted as a key factor in achieving efficiency cost effective and sustainable fuel cell applications.

For hydrogen fuel cells to achieve widespread adoption, innovations in materials, policy support, and infrastructure expansion are essential. With continued investment and technological progress, fuel cells have the potential to drive a cleaner, more sustainable energy future, reducing carbon emissions while offering reliable energy solutions across multiple sectors.

Introduction

The rapid increase in population and progress in human civilization have led to a rise in energy demand, which remains largely reliant on fossil fuels. This in turn brings out more greenhouse gas emissions. Since the greenhouse effect leads to global warming, reducing greenhouse gas emissions and slowing down

climate change are becoming more and more vital. One of the best methods towards this goal is the usage of fuel cell which was invented by W. R. Grove in 1839 (Grove, 1839a).

Fuel cells are electrochemical devices that efficiently transform chemical energy stored in the fuel and oxidant into electricity and heat through redox reactions. Fuel cells are considered as a sustainable technology that not only mitigates environmental challenges, such as excessive local pollution, but also enhances economic benefits through their high efficiency. Fuel cells are recognized as a competitive alternative to combustion engines, because electrochemical reactions are not limited to Carnot's cycle and they can mitigate environmental damage, and the geopolitical issues associated with reliance on fossil fuels (Zaidi & Rauf, 2009). The main advantages of fuel cells are their low maintenance costs, low noise and high energy conversion efficiency (Smitha et al., 2005).

Hydrogen fuel cells, especially the proton exchange membrane fuel cells, not only provide energy with zero emission but also offer much higher conversion efficiency and fast-start up capability (Chandan et al., 2013)(Zhou & Chen, 2023). Even though fuel cells deliver a lot of benefits, they have many challenges and limitations that should be overcome before commercial applications. Catalysts cost and stability (cite), durability, susceptibility to corrosion hinder fuel cell to be commercialized. In addition, high density operations usually leads to issues such as mass transfer, water and gas management and material degradation. Ineffective handling of water and gas within fuel cell may result in issues such as membrane dehydration, inadequate supply of reactants, or excessive water accumulation, all of which can significantly compromise the cell's efficiency and lifespan (Okonkwo et al., 2021) (Jiao & Li, 2011) (Li et al., 2008). In order to avoid those problems and improve power output, optimization strategies are used. Fuel cell efficiency can be enhanced through the optimization of individual components. Despite ongoing advancements, many optimization models for fuel cells still rely on oversimplified assumptions, often overlooking mechanical compression effects, real-world material behaviors, and full-stack variability. Additionally, sustainability challenges such as high emissions from fossil-fuel-based hydrogen and intermittency in renewables highlight the need for balanced solutions. Hybrid fuel cell systems present a promising path forward, and their optimization is essential to maximize efficiency, minimize environmental impact, and ensure scalability for future energy demands.

In this paper an overview of fuel cell and how we can take one more step towards sustainability is discussed. The review starts with an explanation for fuel cell concept and components. The review subsequently provides background details on various types of fuel cells along with their operational characteristics, limitations and challenges. Then, some advancements in fuel cell vehicles and life cycle assessment studies have been discussed. Finally, the review discusses the benefits of fuel cell optimization, highlights certain limitations, and proposes a method to overcome these challenges.

Catalysts

Although platinum is very effective as an electrode and brings about high fuel cell efficiency, its high cost and susceptibility to poisoning make researchers seek for new alternative to reduce its use. (Tasic et al., 2009) showed that even though efficiency is lower, using non-noble metal cobalt (Co) as a catalyst would be competitive with platinum (Pt), as cost efficiency ratio is considerably lower. Moreover, supported catalysts can also be applied to reduce the amount of platinum (Pt) on electrodes (Cho et al., 2007). Furthermore, in 2019, (Van Dao et al., 2019) conducted a study on Pt/C catalyst electrode with double catalyst layer (Pt-dispersed and Pt-concentrated), demonstrating that it has significantly better performance on HOR than Pt/C with single catalyst layer (SCL). Besides that, Pt catalysts are susceptible to carbon monoxide (CO) which leads to fuel cell degradation rapidly. In order to avoid this (Chang et al., 2023) examined synthesized hybrid MXene ($\text{Ti}_3\text{C}_2\text{T}_x$) and carbon nanotube (CNT) loaded Pt nanoparticles and employed as catalytic support. From the results it was observed that after applying 1000 ppm CO and H_2 , electrochemically active surface area (ECSA) loss of Pt/ $\text{Ti}_3\text{C}_2\text{T}_x$ – CNT was considerably more than that of Pt/C due to lower adsorption

and also onset potential was less which indicates its better CO tolerance. On top of that, to tackle this issue, a new strategy was presented by (Tongtong Yang, 2024). Yand et.al and colleges synthesized high density synergistic active sites between iridium single atoms (Ir-SACs) and iridium nanoparticles (Ir-NPs) on a nitrogen-doped carbon nanosheet (NC). This showed significantly superior performance as catalyst in terms of CO tolerance at 1000 ppm CO/H₂ which can be attributed to abundance of single atom and nanoparticle interface and its ability to oxidize CO at low potentials. [Table 1](#) summarizes various catalysts used in different fuel cell types, including their active materials, advantages, and limitations.

Table 1—Catalysts

Catalyst Type	Fuel Cell Type	Active Material(s)	Advantages	Limitations
Platinum (Pt)	PEMFC, AFC, DMFC	Pt nanoparticles, Pt/C	High activity for HOR/ORR; excellent conductivity; durable in acidic environments	High cost; susceptible to CO poisoning; degrades at high current densities
Platinum-Ruthenium (Pt-Ru)	DMFC (anode)	Pt-Ru alloys	Enhanced CO tolerance for methanol oxidation	Ruthenium dissolution over time; high cost
Gold (Au)	AFC, DBFC	Au nanoparticles; Au (100) single crystals	High ORR activity in alkaline environments; corrosion-resistant	Expensive; lower activity than Pt in acidic FCs
Silver (Ag)	AFC, DBFC	Nanoporous Ag; Ag/C	Lower cost than Pt/Au; good stability in alkaline FCs	Moderate ORR activity; forms oxides that reduce performance
Non-PGM Catalysts	PEMFC, AFC	Fe-N-C; Co-N-C; Ni	Low cost; avoids PGM scarcity; stable in alkaline environments	Lower activity than Pt; instability in acidic PEMFCs
Palladium (Pd)	DMFC, DBFC	Pd/C; Pd-Co; Pd-Cu	High methanol tolerance (DMFC); good BH ₄ oxidation (DBFC)	Expensive; carbonate sensitivity (DBFC)
Iridium (Ir)	PEMFC	Ir-SACs/Ir-NPs on N-doped carbon	Exceptional CO tolerance at 1000 ppm; low oxidation potential	Complex synthesis; scalability challenges
Nickel (Ni)	AFC, DBFC	Ni foam; Ni alloys	Low cost; suitable for alkaline environments (AFC/DBFC)	Low faradaic efficiency (50% in DBFC); hydroxide formation degrades anode

Fuel Cells Challenges and Limitations

Fuel cells have been widely recognized as an alternative energy source for a variety of applications, including transportation, stationary power generation, and backup power systems. They offer the potential for high efficiency, low emissions, and sustainability compared to traditional combustion-based power systems. However, despite their advantages, fuel cells still face several limitations and challenges that hinder their widespread adoption.

Fuel cells have the potential to become a key energy source, but addressing their existing challenges is crucial for broader adoption. The main concerns revolve around cost, efficiency, and durability, leading to extensive research on developing more resilient and cost-effective materials.

PEMFC

Proton Exchange Membrane Fuel Cells (PEMFCs) are recognized as a highly promising technology for future applications, particularly in transportation, decentralized power generation, and portable electronic devices (L. Wang et al., 2003). Several critical challenges must be addressed before Proton Exchange Membrane Fuel Cells (PEMFCs) can be commercially viable. One of the primary obstacles is the

development of alternative materials to replace the noble-metal catalyst, particularly platinum (Pt), which is currently utilized for hydrogen dissociation into electrons and protons. Although platinum remains the most efficient catalyst for fuel cell applications, its high cost and susceptibility to carbon monoxide (CO) poisoning significantly hinder its widespread adoption (Nordin, 2010).

Generally, most of problems such as mass transfer, water and gas management, degradation arise at high current density operations. One of the problems in PEMFC is that when current density is high, more proton transfer need is inevitable and as the proton conductivity is low in both membrane and catalyst layer (CL), it in turn, leads to performance degradation. One of the possible solutions to this was to decrease membrane thickness in order to improve proton conductivity and minimize ohmic losses (Nouel & Fedkiw, 1998), however, according to the results obtained from (Yuan et al., 2010)'s study, using thinner membranes would result in fast performance degradation which was attributed to hydrogen crossover due to pinhole formation. Thus, addressing the trade-off between proton transport efficiency and ohmic resistance, while simultaneously mitigating potential durability challenges associated with the proton exchange membrane (PEM) under high current density (HCD) conditions, is of critical importance (Fengyang Cai, 2024). (Xue et al., 2022) highlighted that as the ionomer content increases first ohmic resistance drops (by 30%) while proton diffusion increases and in contrast, at higher ionomer concentrations resistance increases and fuel cell experiences performance loss.

PAFC

One of the main issues is phosphate (H_3PO_4) poisoning which leads to degradation of platinum-based catalysts in PAFC (He et al., 2013). The poisoning effect is attributed to the replacement of a single H_2 molecule with two CO molecules on the platinum surface, occupying dual active sites (Fuel Cell Handbook, 2000). On top of that, platinum catalysts are susceptible to sulfur poisoning at the anode in the presence of impure or reformed hydrogen. This happens mainly due to the blockage of active surface of platinum catalysts for hydrogen oxidation which leads to drop in power output and overall efficiency (O'Hayre et al., 2016). Sulfur poisoning usually happens when H_2S concentration is more than 50 ppm. In case CO is also present together with H_2O , the effect becomes even more severe. Corrosiveness of liquid electrolyte is also another challenge that should not be overlooked. As PAFC is working at high temperature, after certain time, phosphoric acid (H_3PO_4) starts to evaporate, and vapor accumulates which leads to cell corrosion. In this case, phosphoric acid should be replenished. Another factor that results in oxygen reduction reaction degradation is the emergence of phosphate salt ($(\text{NH}_4)\text{H}_2\text{PO}_4$) especially when it is more than 0.2 mol%.

AFC

One of the key advantages of Alkaline Fuel Cells (AFCs) is their ability to utilize a range of non-precious metals as catalysts, significantly lowering costs. However, AFCs are highly susceptible to carbon dioxide (CO_2) contamination, which can disrupt cell performance. Since the amount of carbon dioxide in the air is enough to affect the performance of AFC, pure oxygen should be used to avoid the reaction below:



OH^- comes from KOH electrolyte, produces CO_3^{2-} and reacts with KOH to precipitate K_2CO_3 which in turn degrades AFC. Other factors that the reaction of OH^- and CO_2 hinders AFC performance are that it reduces electrolyte conductivity, limits diffusion rate and oxygen solubility. Therefore, KOH electrolyte should be replenished to partially help this issue. Beyond this, material degradation remains a major challenge to their commercialization. While AFC stacks have demonstrated stable performance for over 8,000 operating hours, large-scale utility applications demand a lifespan of at least 40,000 h. Furthermore, the water generated during hydrogen oxidation reaction (HOR) at the anode must be removed.

MCFC

Molten Carbonate Fuel Cells (MCFCs) require enhanced resistance to sulfur poisoning, especially in coal-based applications. Developing sulfur-tolerant anode materials is essential to eliminate the need for fuel cleanup systems, which reduce efficiency and increase costs. This challenge is more critical when low-temperature desulfurization is required, as it results in energy losses by cooling and then reheating the fuel gas to MCFC operating temperatures (Fuel Cell Handbook, 2000). In addition, since bipolar plates and current collectors are mainly made of stainless steel, these components suffer from corrosion or chromium and iron oxidation, leading to metal ion contamination in the electrolyte.

Another challenge is carbon deposition which can lead to plugging of gas channels by accumulating in gas flow lines. When pure cells are run on CO, carbon deposition can occur through following reaction:



It usually happens in case of CO and CO₂ disproportionation. However, if the rate of 2CO₂/1CO is maintained, carbon deposition can be prevented (Webb et al., 1965). In addition to that, carbon deposition can occur even when pure hydrogen is utilized as fuel.

Fuel Cells in Transportation

As the amount of greenhouse gas emissions grows rapidly in the atmosphere, the need for alternative energy and non-fossil fuel vehicles is also increasing. Considering that transportation industry accounts for 17 % of annual greenhouse gas emissions (Naef A. A. Qasem, 2024), recently governments are more prone to focusing on electric vehicles (EVs) and fuel cell electric vehicles (FCEVs) since they are the best candidates for long-term replacement for internal combustion engine vehicles (ICEV). Fuel cells have the capability to be used in diverse modes of transportation such as buses, motorcycles, ships, aircrafts, trains, and trucks (Andújar & Segura, 2009). Other advantages of FCEVs are its quietness and higher energy conversion efficiency (Fakhreddine et al., 2023). In addition, hydrogen has significantly higher energy density per kilogram than hydrocarbons. FCEVs demonstrate superior range capabilities and faster refueling processes when compared to battery electric vehicles (BEVs). That's why there has been a significant increase in the number of FCEVs over the last decade. In 2022, there were 72,200 FCEVs (Outlook, Global EV Outlook, 2023), and this figure rose by approximately 21% in 2023, reaching 87,600 (Outlook, Global EV Outlook, 2023). World's first FCEV, more specifically Electrovan, was produced by General Motors in 1966 ("GM Marks 50 Years of FCEV Development, from Electrovan to Chevrolet Colorado ZH2," 2016).

China had the largest number of stations in 2023, at over 400, followed by Europe, with 280, Korea at 180, and Japan with over 170 (Agency, 2024). China government even invested to reach over 1 million FCEV by 2030 (Yeung & Liu, 2023). Hyundai, Toyota and Honda are the leading brands in FCEV sales, dominating the market with their advanced hydrogen technology (Aminudin et al., 2023).

Fuel cells deliver performance equivalent to traditional internal combustion and diesel engines but show enhanced fuel efficiency especially when they operate with synthetic fuels made from coal. Methanol stands out as an excellent fuel source because it can be transformed into hydrogen that combines with oxygen within an acid electrolyte fuel cell to generate electricity (Gonzalez & Srinivasan, 1984).

Buses

Urban areas across the globe are embracing fuel cell buses as a cleaner substitute for traditional diesel-powered public transportation systems. In the past, in most applications, hydrogen fuel cells were being supplied with the hydrogen obtained from methanol reforming. However, steam reformers were also placed in the car where hydrogen was obtained by filling methanol. For example, in 1997, Georgetown University developed a 30-foot hybrid-electric fuel cell transit bus powered by the system with 155 kW output power (55kW-PAFC and 100 kW-battery pack) with on-board reformer (Wimmer, 1997).

One of the main projects, the so-called Zero Emission Bay Area (ZEBA) Fuel Cell Bus Demonstration, is run by Alameda-Contra Costa Transit District (AC Transit) with 13 advanced-design fuel cell buses. According to the report (2017) from National Renewable Energy Laboratory (NREL), the economy of fuel cell electric buses (FCEBs) is 45 % more than that of diesel buses in terms of miles per diesel gallon equivalent (Leslie Eudy, 2017).

Passenger vehicles

Proton exchange membrane fuel cells (PEMFCs) are typically used in passenger vehicles due to their ability to start up quickly and operate at low temperature ($\sim 80^{\circ}\text{C}$) and high power density (Fakhreddine et al., 2023). In addition, PEMFCs have good tolerance to CO_2 and its electrolyte is not corrosive. Therefore, PEMFC is considered the most appropriate fuel cell type for passenger vehicles (Panik, 1998) (Aminudin et al., 2023). As an example, Hyundai introduced Tucson ix 35 fuel cell car with 100kw power and with a capacity of 27.8 km/ liter H_2 which is significantly higher than the fuel economy of gasoline (11.2 km/liter). Tucson ix35 had 415 km driving range with 80,000 km durability. However, they have substantially improved fuel cell vehicle technology in the new version, so-called Nexa, increasing the driving range to 609 km and durability to 10 years (160,000 km) durability (Hong & Kim, 2018). Moreover, Toyota commercialized Toyota Mirai hydrogen fuel cell vehicle (FCV) that uses Toyota Fuel Cell System (TFCS) which is more efficient than internal combustion engines.

Though the number of fuel cell passenger cars was limited to 11,200 in 2018 (Outlook, Global EV Outlook, 2019), it had climbed to nearly 57,400 in 2022 (Outlook, Global EV Outlook, 2023) and around 66 065 in 2023 (Agency, 2024).

Motorcycle

On top of that, there have been a few fuel cell motorcycle projects in the last two decades. For example, Pios fuel cell motorcycle with 1.3kg hydrogen storage capacity and 300 kg range (J. Weigl et al., 2008) and H2Motive® fuel cell motorcycle with 9.4 hp net power (fuel cell) by UTM team (J. D. Weigl et al., 2016).

Trains

Hydrogen fuel cell electric trains offer several advantages over electric trains, including reduced life-cycle emissions, shorter refueling durations compared to battery-electric trains. In addition, the electric trains that are powered by solar energy can have energy shortages during nighttime hours, therefore, adopting hydrogen trains in this case can have great benefits. Furthermore, unlike other electric trains, hydrogen fuel cell electric trains do not require overhead electric wires (Ding & Wu, 2024). The world's first hydrogen-powered urban train was developed by CRRC Changchun Railway Vehicles that can run at a speed of 160 km/h (CRRC, 2023). In 2024, the initiation of Hymпульso project was announced by Talgo in Spain. In this project it is planned to build the world's first hydrogen-powered high speed train that can run at a speed of 300 km/h (Talgo, 2024). "The Calgary Airport – Banff Rail (CABR)" project is a proposed hydrogen-powered rail service designed to connect Calgary International Airport to Banff, offering a seamless, eco-friendly "one-seat ride" that aims to reduce traffic congestion and emissions (CABR, 2020).

Challenges

Hydrogen refueling stations

One of the main reasons delaying widespread commercialization of hydrogen fuel cell vehicles is limited availability of hydrogen refueling stations (HRSs). However, the number of hydrogen refueling stations are growing rapidly in the world. For example, in 2018 there were only 381 (Outlook, Global EV Outlook, 2019) refueling points, but it increased to almost 1,200 in 2024 (Agency, 2024). Despite this rapid growth

(Figure 1) Growth in Hydrogen Refueling Stations (HRS), achieving a sufficient ratio of FCVs to HRSs still requires significant capital investment and time.

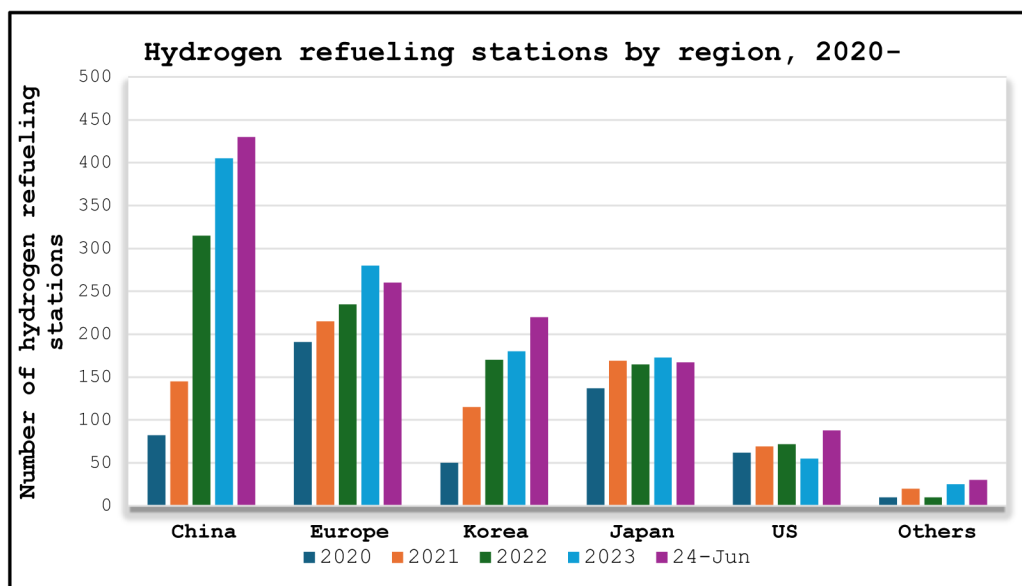


Figure 1—Growth in Hydrogen Refueling Stations (HRS) (Agency, 2024)

Hydrogen production

Even though hydrogen fuel cells do not produce greenhouse gases, the most common hydrogen production method is steam methane reforming (SMR), which releases CO₂ (Yukesh Kannah et al., 2021). To make hydrogen fuel cells vehicles 100% emission-free, hydrogen production from water electrolysis powered by renewable energy is the best option. However, it is costly and faces challenges to meet large-scale demand. In fact, the average cost of renewable-based green hydrogen can be three to five times more than the energy equivalent of natural gas or oil, primarily due to the high electricity consumption required for electrolysis and the limited scale of production facilities Figure 2 (Shiva Kumar & Himabindu, 2019) (Megia et al., 2021). While fossil fuels are extracted, refined, and distributed through well-established global supply chains, hydrogen still requires expensive infrastructure for compression, liquefaction, and storage. This economic disparity makes hydrogen less competitive today, even though its environmental benefits position it as a critical fuel for the long-term energy transition.

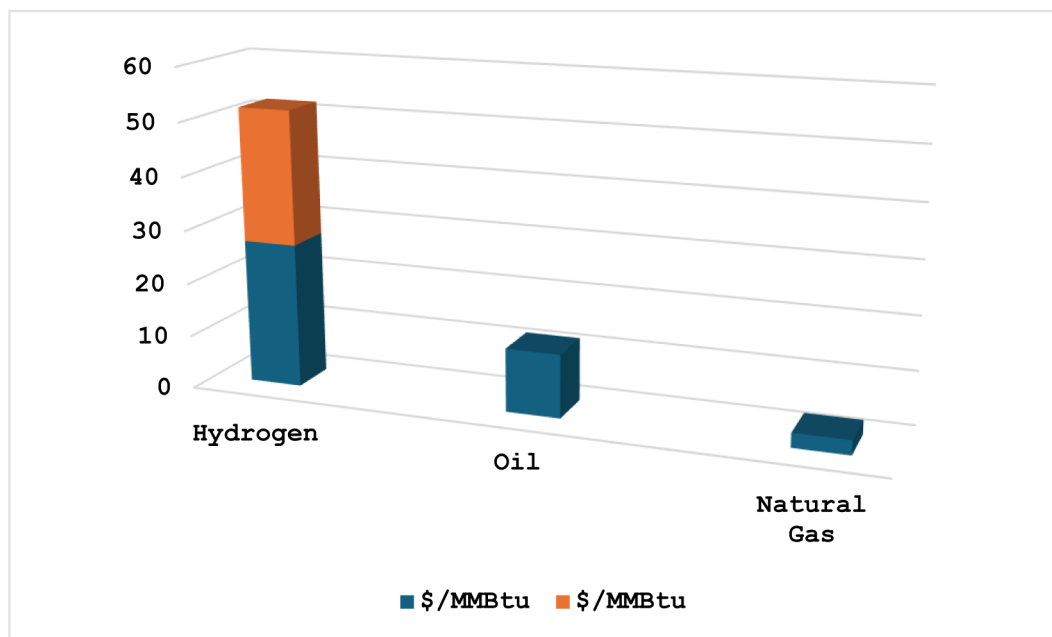


Figure 2—Comparison of the prices of hydrogen and fossil fuels

Energy consumptions and Prices

The manufacturing cost of FCEVs is higher than BEVs and commercial ICEVs. Fuel cells require costly catalysts, special type of storage tanks, because as hydrogen has very small molecules, they diffuse into metal structures and cause material failure and brittleness, which in turn adds to the cost. To overcome this challenge, there is a demand for specific coatings that are very expensive. For instance, production of 40-ft fuel cell bus costs \$1.5 - \$2.0 million which is around 5 times more expensive than equivalent diesel bus (Garche & Jörissen, 2015). In addition to that, despite the fact that hydrogen has higher gravimetric energy density, hydrogen fuel is more expensive than fossil fuels, because of the high energy consumption for compression, liquification and transportation. Liquification of hydrogen is very energy-intensive process, because hydrogen gas must be cooled down to -253°C at high pressure.

Furthermore, membrane electrode assemble (MEA) and catalysts degrade over time due to mechanical stress, chemical reactions, and impurities in hydrogen fuel. Consequently, all these factors contribute to higher prices and temporarily hinder rapid commercialization of HFCVs or FCEVs.

Consequently, from the rapid growth of the number of vehicles and hydrogen refueling stations, it can be deduced that hydrogen has significant potential for widespread adoption in the near future. Continuous improvements in hydrogen energy - including fuel cell vehicles, refueling stations and being aware of these numbers leads to greater public acceptance, thus faster development in the hydrogen energy sector (Hienuki et al., 2019) (Q. Wang et al., 2020).

Hydrogen Fuel Cell Optimization

The main obstacle to the widespread commercialization of fuel cells lies in achieving their maximum performance while minimizing overall costs. This challenge stems from the need to balance multiple competing factors, including material expenses, operational efficiency, system durability, and manufacturing complexity. Optimization of fuel cells, therefore, has emerged as a critical approach to address these trade-offs and enhance their practical viability. An optimized fuel cell must not only deliver the highest possible power output and energy efficiency but also ensure long-term durability, reliability under variable operating conditions, and minimal degradation over time. At the same time, it must be cost-effective, requiring innovations in material selection, structural design, and integration techniques.

Optimization efforts typically involve multi-objective analyses that consider thermal management, water transport, reactant distribution, and electrochemical performance, often supported by numerical modeling and experimental validation.

Flow field

Fuel cell performance can be evaluated based on several factors such as polarization curves, water, thermal states and power density curves. (Jiang et al., 2024) examined 5 different streamlined blockage design for flow channel of the PEMFC and evaluated energy output. In addition, they found the best option between them by implementing optimization studies. Optimization case of streamlined block design was determined by TOPSIS (Technique for Order Preference by Similarity to Ideal Solution). It was concluded that an elliptical blockage yielded the highest increase in electrochemical conversion efficiency (about 3.42 %). (Ahmed & Sung, 2006) evaluated three different channel cross-section shapes (rectangular, trapezoidal, parallelogram) using 3D non-isothermal model. They reported that even though rectangular channel cross-section resulted in the highest cell performance, more uniform and local current density were obtained with trapezoidal shape. Additionally, the optimum channel-shoulder width ratio was found to be 1.3-1.4. (C. Yang et al., 2021) performed numerical simulations to investigate the effect of M-shaped flow field with different height and width of obstacle on fuel cell performance. According to their findings, fuel cell performed at highest level when height and width was 0.15 mm and 0.40 mm, respectively. Then, the opposite sinusoidal wave flow channel was proposed by (Zhou et al., 2022). They developed 3D two-phase CFD models of flow channels with different amplitudes and periods to assess the performance of PEMFC. It was concluded that the opposite sinusoidal wave flow channel remove water faster and improves net power density by 16.61 % at 2A/cm². (Najmi et al., 2021) also studied the implications of non-straight parallel flow field and reported that the new wavy channel design performs better in comparison because it helps to reduce ohmic resistance and improve mass transfer.

One of the studies that also intended to improve fuel cell performance more specifically, uniformity in the distribution of reactant species and current density over the active area of the fuel cell has been done by (Carcadea et al., 2011). In this work, to assess how changes in permeability of porous layer (GDL and CL) affect fuel cell performance, numerical simulations were carried out using a 2-phase model across 4 different permeability values ranging from 10⁻⁹ to 10⁻¹² m². The findings indicate that average current density was highest (around 0.86 A.cm²) at lower permeability values of 10⁻¹², 10⁻¹¹, and 10⁻¹⁰ m². Furthermore,

Apart from these, other flow field characteristics such as cross-sections shape, horizontal and vertical structure, shape, arrangements, and structure of baffles, can also be manipulated to optimize and improve the performance of fuel cells (Zhou & Chen, 2023).

Beyond flow fields, the material properties and dimensions of fuel cell components such as electrodes, catalyst layers, gas diffusion layers, membrane critically determine performance. Researchers have used modeling and simulation to identify optimal thicknesses, porosities, and compositions that maximize reactions while minimizing material usage which is important for cost reduction.

GDL and CL

In PEMFC, the catalyst layer (CL) thickness influences both performance and cost, since thicker layers allow more reaction sites but also use more precious catalyst, thus increasing cost and can lead to mass transport limitations. (Xia, Ni, Xu, et al., 2021) performed a numerical non-isothermal 3D modeling study to find the optimal CL thickness for a high-temperature PEMFC at 160°C and 0.6V. They initially identified the effective reaction thickness (ERT) to have a trade-off between CL thickness and electrochemical performance. Given the insignificant effect of anodic CL thickness on ERT, the study focused on cathodic CL thickness, examining a range of 5 µm to 100 µm. They concluded that in order to keep the cost minimum and maximize the performance the most appropriate CL thickness ranges are 10 to 17 µm and 15 to 30 µm.

Another study was implemented by (Huang et al., 2022) on the optimization of CL thicknesses in anode and cathode separately via genetic algorithm in MATLAB and COMSOL for HT-PEMFC. They initially created 3D non-isothermal U-shaped serpentine flow field (Figure 3) model. The optimal values of CL thicknesses in anode and cathode were found to be 0.0231 mm and 0.0315 mm correspondingly which led to 6.5% growth in fuel cell performance.

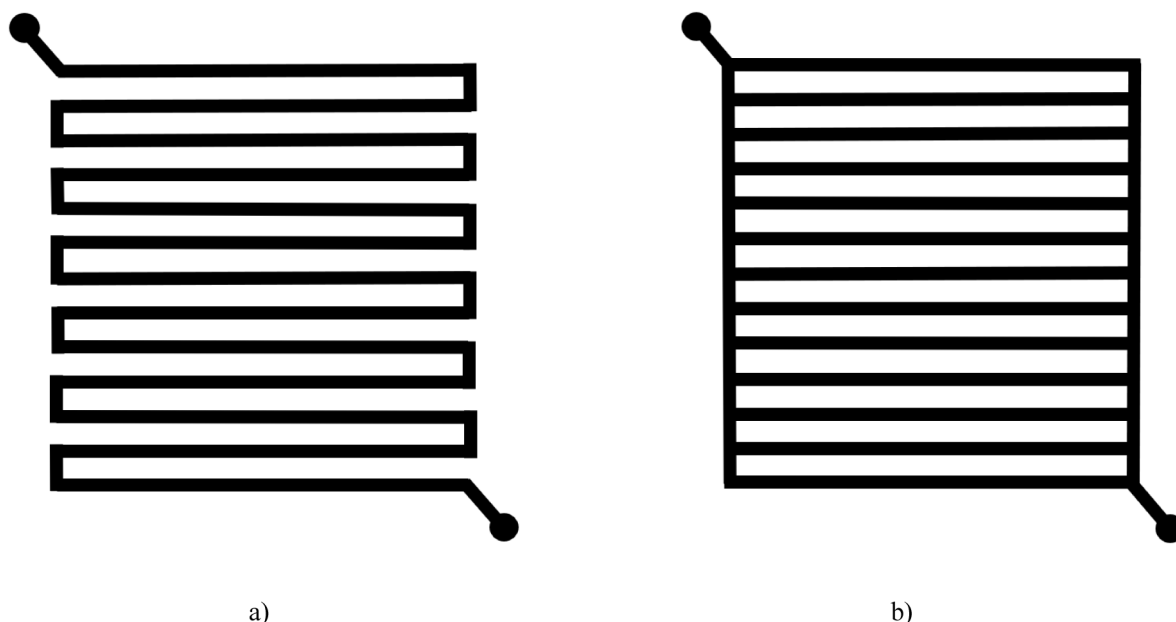


Figure 3—Flow field designs of PEM fuel cell channels a) Serpentine b) Parallel

(Kamarajugadda & Mazumder, 2008) developed a two-dimensional CFD model for PEMFC to investigate the impact of cathode catalyst layer structural and compositional parameters on PEM fuel cell performance. The results showed that 0.45 mg/cm² is the most efficient value for platinum mass (mPt) in order to have enough catalyst sited for high reaction rates alongside avoiding excessive thickness that harms oxygen transport and increase cost. They also obtained polarization curves and concluded that platinum-carbon mass ratio (Pt/C) should be 0.28 for the optimum performance. Moreover, an optimum value of 60 nm of polymer coating thickness was suggested.

(Xia, Ni, He, et al., 2021) investigated the optimal GDL properties (thickness and porosity) through a numerical non-isothermal three-channels 3D model was validated with experimental data. GDL optimization improves reactant distribution and cell performance, especially at the cathode, with initial thickness increases enhancing oxygen flow and average performance. However, excessive GDL thickness or porosity eventually hinders performance due to increased ohmic resistance and decreased electron conductivity, despite improved flow uniformity. Therefore, the optimal GDL thickness was found about 80–120 μm on the anode side and 140–170 μm on the cathode side, with an optimal porosity of 35–45%. However, they did not consider compression effect during fuel cell assembly.

Another factor that affects the performance of fuel cell and hinders its commercialization to some extent is the nonuniform distribution of reactants in fuel cells which results in the degradation of catalyst. Therefore, fuel cells' lifespan decreases considerably. This can be alleviated by the optimization of GDL porosity distribution which is important for improving mass and water management (Zhang et al., 2024). Zhang and colleagues studied PEMFC performance deeply via modeling 3 different porosity distribution such as the uniform, curvilinear, and parabolic distribution. After optimization with the coordinate rotation method, it was deduced that the most advanced uniform distribution of PEMFC current density can be obtained through parabolic distribution of GDL porosity. In addition, (Ebrahimi et al., 2017) optimized the catalyst

loading profile while keeping the total amount of catalyst constant. They employed 2D steady-state CFD model of cathode side and GA algorithm with the objective of maximizing power density. They found out that uniform catalyst loading is not efficient because oxygen is abundant at the channel entrance but depleted near the exit. So, they optimized catalyst loading with 2.3 times more catalyst near the inlet compared to the outlet, because placing more catalyst where there's less oxygen (near the outlet) is wasteful. The results showed that power density improved by 14% without increasing Pt usage. However, 2D simplification assumes uniform contact, ignoring non-uniform compression.

Furthermore, (Zheng, 2024) implemented the first study to introduce individual optimization of porosity and permeability across distinct regions of the GDL by applying topology algorithms. They applied the topology algorithm to individual small GDL units within each section, using initial porosity values of 0.4 and 1 as the starting parameters and then extended to each section of GDL. They concluded that when only porosity is taken into consideration and set to 1 at 0.6V, the performance increased by 10.3%, whereas this was 5.3% when both permeability and porosity were optimized together. However, the model oversimplifies the complex three-dimensional nature of transport phenomena and non-uniform compression in real GDL structures.

Table 2—Summary of fuel cell system optimization studies

Optimization target	Method/Model/Software	Optimized value/Outcome	Reference
Cathodic CL thickness	A numerical non-isothermal 3D modeling	10 to 17 μm and 15 to 30 μm	(Xia, Ni, Xu, et al., 2021)
CL thickness	Genetic algorithm in MATLAB, COMSOL 3D non-isothermal U-shaped serpentine flow field model	23.1 μm (anode) 31.5 μm (cathode) 6.5% higher η	(Huang et al., 2022)
Streamlined blockage design	TOPSIS	Elliptical (3.42% higher η)	(Jiang et al., 2024)
GDL porosity distribution	3D two-phase isothermal geometric model, COMSOL Multiphysics 5.5	Parabolic distribution performs better	(Zhang et al., 2024)
GDL thickness Porosity	A numerical non-isothermal three-channels 3D model	80–120 μm (anode side) 140–170 μm (cathode side) 35–45% (7.7% higher cell performance)	(Xia, Ni, He, et al., 2021)
Flow field design	3D CFD	Serpentine performed than parallel	(Ferng & Su, 2007)
Porous layer (GDL and CL) permeability	2 phase numerical modeling	0.86 (max) current density from 10^{-12} to 10^{-10} m^2	(Carcadea et al., 2011)
Bipolar plate channel geometry	2D and 3D CFD with GA algorithm	Normal: channel-to-rib ratio = 2.8:0.5 Reverse: channel-to-rib ratio = 4.2:0.3	(W. J. Yang et al., 2015)
Platinum mass Platinum-carbon mass ratio Polymer coating thickness	2D CFD	mPt = 0.45 mg/cm ² Pt/C = 0.28 Thickness = 60 nm	(Kamarajugadda & Mazumder, 2008)
Catalyst distribution	2D steady-state CFD model with GA algorithm (C++)	m _p (min) = 0.6 m _p (max), 2.3 times more catalyst in inlet (14% higher cell performance)	(Ebrahimi et al., 2017)
Porosity Permeability	Density gradient-based topology algorithm	Average porosity = 0.56 (10.3% higher cell performance) Average permeability = $5.2 \times 10^{-13} \text{ m}^2$	(Zheng, 2024)
M-shaped Flow Field with different height and width of obstacle	3D SIMP algorithm FLUENT	H = 0.15 mm W = 0.40 mm	(C. Yang et al., 2021)

Hybrid Fuel Cell – Can it be the Key Step Towards Commercialization?

The commercialization of fuel cells depends on achieving high performance at low cost through optimization of materials, geometry, and system integration. Flow field designs such as serpentine or elliptical channels significantly improve mass transfer, reactant distribution, and thermal regulation, while optimization of gas diffusion and catalyst layers via porosity engineering, graded structures, and spatial catalyst loading enhances water management and power density without raising platinum use.

However, most models remain idealized, often neglecting real-world factors like mechanical compression, durability under variable loads, and anisotropic material behavior, limiting scalability to full-stack systems. Sustainability also plays a critical role: hydrogen from fossil-based routes like SMR offers efficiency but produces high emissions, whereas renewable-based electrolysis is cleaner but costlier and less consistent.

A promising pathway is hybrid fuel cells, integrating renewable and fossil energy to balance environmental and economic benefits. Yet, studies on hybrid system optimization are scarce, highlighting the need for research on materials, lifecycle impacts, and economic feasibility to enable large-scale deployment.

Conclusion

The advancement of fuel cell technology has been a long journey, from its conceptualization in the 19th century to its present-day applications in transportation, power generation, and other industries. Hydrogen fuel cells, in particular, have emerged as a promising alternative to conventional energy sources, offering high efficiency, zero emissions, and long-term sustainability. However, despite these advantages, the widespread commercialization of fuel cells faces several challenges, including cost, infrastructure, durability, and hydrogen production methods.

Key Challenges in Commercialization

The main barrier is cost, driven by platinum catalysts and complex stack manufacturing. Innovations in materials, production methods, and scaling are vital to competitiveness. Optimization of components such as flow fields, gas diffusion layers (GDL), and catalyst layers (CL) has been shown to improve water management, current density, and power output while reducing costs.

Infrastructure is another bottleneck, as hydrogen refueling networks lag far behind those of gasoline and electricity. Expanding production, storage, and distribution capacity is essential for fuel cell vehicle adoption. Hydrogen sourcing is also critical: most is produced via carbon-intensive steam methane reforming (SMR), while green hydrogen from renewables remains costly but is expected to decline in price with electrolyzer advances and falling renewable costs. Durability concerns, including catalyst degradation and membrane fouling, further limit long-term performance.

Future Outlook and Path Forward

The path forward lies in hybrid systems that balance fossil reliability with renewable benefits, supported by optimized materials, system design, and lifecycle strategies. Advances such as non-precious metal catalysts, high-temperature membranes, and improved bipolar plates will reduce costs, improve efficiency, and accelerate fuel cells toward sustainable large-scale commercialization.

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